

EDUCATION

Present Sept. 2025	PSL University MSc in Mathematics	Paris, France GPA: 15.052/20
	Coursework at Dauphine and École Normale Supérieure. - Ranked 5th of 116 students, top 5%. - Full scholarship awarded by the Fondation Sciences Mathématiques de Paris.	
June 2025 Sept. 2020	University of Barcelona BSc in Mathematics	Barcelona, Spain GPA: 9.02/10
	- Ranked 5th of 141 students, top 4%. Honors in 12 subjects. - Thesis on flat connections and principal bundles. ↗ Grade: 10/10	
June 2025 Sept. 2020	University of Barcelona BSc in Physics	Barcelona, Spain GPA: 8.79/10
	- Ranked 16th of 196 students, top 8%. Honors in 10 subjects. - Thesis on classical shadows and quantum computing. ↗ Grade: 9.7/10	

WORK EXPERIENCE

Present July 2025	Westinghouse Nuclear <i>Simulation Modeling Engineer</i>	Madrid, Spain
	- Develop Fortran-based simulation software for nuclear power plant thermal-hydraulic systems. - Extend existing models from single-phase to multiphase flow. - Validate simulation outputs against engineering benchmarks and document model assumptions for technical review.	

RESEARCH EXPERIENCE

June 2025 Nov. 2024	Barcelona Supercomputing Center – BSC <i>Visiting Student</i>	Barcelona, Spain
	- Implemented a Shadow-Grouping algorithm combining classical shadows with observable grouping to improve quantum measurement efficiency. - Benchmarked the method on H₂ and LiH Hamiltonians using PennyLane. - Improved energy-estimation accuracy by up to 18-fold with two to three orders of magnitude fewer measurements than standard classical shadows. - Supervisors: Prof. Dr. Bruno Julià, Dr. Berta Casas, Dr. Sergi Masot.	
Feb. 2025 July 2024	Institute for Research in Biomedicine – IRB <i>Research Intern</i>	Barcelona, Spain
	- Modeled DNA flexibility using coarse-grained computational methods in C. - Extended the model to analyze how DNA damage affects flexibility. - Developed Python algorithms to compute DNA geometric descriptors. - Supervisor: Prof. Dr. Modesto Orozco.	

Aug. 2023
June 2023

- Studied bifurcation phenomena in the complex exponential family, focusing on the non-structural stability of $\exp(z)$.
- Supervisors: Prof. Dr. Núria Fagella, Prof. Dr. Xavier Jarque.

HONORS AND AWARDS

- **2025** FSMP Scholarship for postgraduate studies at PSL, among more than 1,000 candidates.
- **2024** Collaboration Scholarship for research toward the Mathematics BSc thesis.
- **2024** Maths4Life Award, one of five IRB grants in mathematics and biomedicine.
- **2024** CRM Award for a Bachelor's thesis supervised by a CRM-affiliated researcher; declined.
- **2021–2025** Excellent with Honors in 16 subjects across the two BSc programs at UB.
- **2020** Honors in Baccalaureate for outstanding academic performance in high school.
- **2019** Mad for Science Programme, selective Physics course introducing students to research.

SELECTED PRESENTATIONS

- **2025** Research Seminar at BSC: presented thesis work on Shadow-Grouping for measurement-efficient quantum energy estimation.
- **2024** Extraordinary BioMed Seminar – Maths4Life: presented research on coarse-grained modeling of DNA flexibility.
- **2023** Barcelona Introduction to Mathematical Research: presented work on bifurcation phenomena in the complex exponential family.

LANGUAGES

Spanish Native
Catalan Native
English C1
French A2

CODING SKILLS

Proficient Python, C, Fortran, LaTeX.
Familiar R, Bash, Mathematica, SageMath, HTML, CSS.
Libraries NumPy, SciPy, Pandas, Scikit-learn, Matplotlib, PennyLane.
Tools Git/GitHub, Unix/Linux, Jupyter.